

E. Product Oriented Recurrence

time limit per test: 1 second
memory limit per test: 256 megabytes

Let $f_x = c^{2x-6} \cdot f_{x-1} \cdot f_{x-2} \cdot f_{x-3}$ for $x \geq 4$.
You have given integers n, f_1, f_2, f_3 , and c . Find $f_n \bmod (10^9 + 7)$.

Input
The only line contains five integers n, f_1, f_2, f_3 , and c ($4 \leq n \leq 10^{18}, 1 \leq f_1, f_2, f_3, c \leq 10^9$).

Output
Print $f_n \bmod (10^9 + 7)$.

Examples

input

5 1 2 5 3

output

72900

input

17 97 41 37 11

output

317451037

Note
In the first example, $f_4 = 90, f_5 = 72900$.
In the second example, $f_{17} \approx 2.28 \times 10^{29587}$.

Codeforces Round 566 (Div. 2)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, ms)

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
336375496	Aug/31/2025 15:06	Happy New Year!

→ Problem tags

dp math matrices number theory

*2300

No tag edit access

→ Contest materials

Announcement (en)

Tutorial (en)